

DONGHYEUN LEE

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SUMMARY

Metrology Process Engineer with an M.S. in Chemistry and hands-on experience in **imaging-based metrology, statistical data analysis, calibration, and yield improvement** for semiconductor-adjacent and advanced materials systems. Strong background in **Optical imaging, SEM/AFM, spectrometry, dimensional mapping, and RCA**, with proven ability to translate measurement data into actionable process feedback in cleanroom/manufacturing environments.
(Interview Finalist: Samsung, ASML, Otsuka | Offer: SK Battery America)

CORE SKILLS

Process & Metrology Monitoring Statistical Methods & Data Analysis Technical Presentations SEM (imaging optimization, defect analysis), Defect classification, root-cause analysis, process feedback loops AFM, ellipsometry, UV-Vis, optical microscopy

PROFESSIONAL EXPERIENCE

Senior Researcher – Proteina Inc. | Seoul, Korea | 2021 – 2023

- Designed and optimized **wet cleaning and surface preparation processes**, including **piranha-based organic removal**, to ensure reproducible surface activation prior to functionalization
- Supported **metrology and inspection processes** using **TIRF and optical imaging systems**, performing **dimensional mapping and calibration verification** with **Cr-grid reference materials** to ensure measurement accuracy and data quality
- Built **statistical data pipelines** (Python/MATLAB) to analyze imaging uniformity, signal drift, and repeatability, improving long-term measurement stability
- Established **fast inspection workflows** integrating **AFM, optical microscopy, plate readers, UV-Vis spectrometry, and ellipsometry** for cross-validation and rapid root-cause analysis
- Supported **atomic- and nano-scale surface layer control**, correlating treatment conditions with thickness, uniformity, and optical response and applied SPC and statistical analysis to assess **process drift, repeatability, and measurement R&R** for surface treatment steps

Research Engineer – Palogen LLC | Seoul, Korea | 2020–2021

- Supported **semiconductor-adjacent sensor manufacturing and reliability analysis** in collaboration with external partners and investigated **sensor defects/signal instability/yield loss**,
- Strengthened experience operating in **time-critical, quality-driven manufacturing environments**

Graduate Researcher – Sogang University (M.S., Chemistry) | Seoul, Korea | 2018–2020

- Generated datasets from **micro/nano-fabricated polymer and surface-engineered systems**
- Conducted **optical signal analysis and feature extraction** using **ImageJ, MATLAB, Python**
- Fabricated micro/nano-scale structures using photolithography, resist patterning, plasma surface treatment, and wet chemical processing

Offer Accepted – Manufacturing Innovation Engineer

SK Battery America (SKBA) | Commerce, GA | *Start: TBA*

- Selected for hybrid program focused on **SPC/FDC pipelines, defect detection, and AI-based yield analysis** in battery electrode production

EDUCATION

Sogang University — M.S., Chemistry | GPA: 4.3 / 4.5 | 2018–2020

Hong Kong University of Science & Technology (HKUST) — B.S., Chemical & Bioproduct Engineering | 2010–2013

CERTIFICATIONS

AI & Machine Learning (PGP) — University of Texas at Austin, 2025

Full-Stack Java Development — HiMedia Academy, 2024

Semiconductor Training Certificate — Seoul National University ISRC, 2010